SMARTROUTE

Urban Emergency Road Routing System – Project Requirements

Domain	Civil Engineering and Urban Planning
SDG	11 – Sustainable Cities and Communities
Topic	Single Source Shortest Path (SSSP)

1. Functional Requirements

1. Represent the city road network as a weighted graph. Intersections and important locations are vertices, while roads are edges.

2. Store the distance (kilometres) and estimated travel time (minutes) for each road.

3. Use Dijkstra's algorithm with a priority queue to calculate the route with the minimum estimated travel time.

4. Display the selected route, total distance and estimated travel time.

5. Allow a road to be closed by deleting its edge from the graph.

6. Allow a closed road to be reopened by adding its edge back to the graph.

7. Support one-way and two-way roads using directed and undirected edges.

8. Recalculate the fastest route after a road is closed or reopened.

9. Display a clear message when no route is available between the selected locations.

10. Validate user input, including invalid location numbers and menu choices.

2. Technical Requirements

1. Programming language: Java.

2. Graph representation: adjacency list.

3. Data structures: HashMap, ArrayList and PriorityQueue.

4. Primary algorithm: Dijkstra's shortest path algorithm.

5. Optional comparison algorithm: Bellman–Ford, if implemented.

6. Development environment: Visual Studio Code, IntelliJ IDEA or Eclipse.

7. Output: console-based road network, route, distance and travel-time information.

3. Non-Functional Requirements

1. Efficiency: Calculate routes efficiently for the size of the road network used in the project.

2. Reliability: Handle road closures, reopening and invalid inputs without unexpected program failure.

3. Maintainability: Organise the program into clear classes and methods.

4. Usability: Provide a simple, menu-driven interface.

5. Accuracy: Select the route with the lowest estimated travel time based on the supplied road data.

4. Hardware and Software Requirements

Hardware	Software
Laptop or desktop computer	Java Development Kit (JDK)
Minimum 4 GB RAM recommended	Windows, Linux or macOS
Keyboard and monitor	VS Code / IntelliJ IDEA / Eclipse
—	Java Extension Pack (if using VS Code)

5. Scope and Limitations

The current project is a console-based prototype that uses predefined road data. It demonstrates graph modelling, fastest-route calculation, road closure and reopening, and one-way road handling. Real-time GPS navigation, live traffic updates, automatic road-closure detection and a graphical map interface are future enhancements; they are not part of the current prototype.

6. Expected Outcome

The system should identify the fastest available route between a source and destination, show its total distance and estimated travel time, and find an alternative route when a road becomes unavailable. This supports the broader goal of safer and more efficient urban emergency response.